

Methods for All-Optical Wavelength Conversion

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Abstract: All-optical wavelength conversion is an essential function in contention resolution and packet switching in optical communications. In these applications, we want to convert wavelengths efficiently and quickly. Therefore, we convert wavelengths in the optical domain, taking advantage of nonlinear methods to create efficient and fast techniques. I will discuss three all-optical wavelength conversion techniques, cross-gain modulation (XGM), cross-phase modulation (XPM), and four-wave mixing (FWM). All of these methods take advantage of nonlinear methods, which normally are undesired. I will discuss their performance and show that they are fast and efficient.

Introduction

As future optical communications networks carry more bandwidth, it will become more important to solve the challenges of contention resolution and packet switching. Wavelength conversion is an essential function of contention resolution and packet switching. We want to convert from one wavelength to another, quickly and efficiently. Therefore, we desire all-optical wavelength conversion, which is conversion of wavelengths entirely in the optical domain. This is because the alternative, converting in the electrical domain, is much slower and requires more components, which is costlier. **Instead, what makes all-optical wavelength conversion so great is that it takes advantage of nonlinear phenomenon to convert from one wavelength to another, which proves to be very fast and efficient.** Normally, nonlinear effects are undesired because they impede the quality of a signal, but these effects can be controlled to create wavelength conversion.

All 3 methods I will discuss use a component called a semiconductor optical amplifier (SOA). This component amplifies an optical signal, but what is interesting about them is that they have nonlinear effects. We take employ these nonlinear effects in the 3 methods I will discuss.

The first two methods I will discuss, cross-gain modulation (XGM) and cross-phase modulation (XPM) use the same nonlinear effect in the SOA. When the gain of a SOA becomes too high, after some threshold, the carriers inside the semiconductor become saturated, so the gain falls drastically. This is called carrier depletion and is normally undesired.

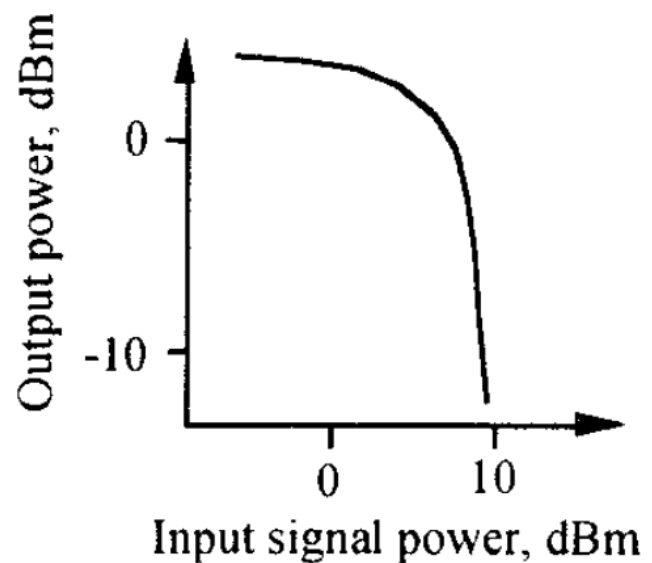


Figure 1: output characteristics of a semiconductor optical amplifier. At some threshold the input power, the output gain drastically falls due to carrier depletion.

In Figure 1, after some threshold input power, the output gain of the SOA falls drastically [1]. As we will see, XGM and XPM takes advantage of this nonlinear effect for wavelength conversion.

In a SOA, when two inputs signals are sent together, the SOA produces beatings, or additional signals at different frequencies. This nonlinear effect is called four-wave mixing, and usually is not desired in optical communications because it degrades performance. But we will see that we can employ this nonlinear effect and control it so that we can produce a signal at our desired wavelength for wavelength conversion.

After discussing these three methods for wavelength conversion, I will discuss the pros and cons of each, comparing their performance. What's

important here is that each type is widely used in the world of optical communications, and usage depends on what kind of network architecture you want to build.

Cross-Gain Modulation (XGM)



Figure 2: Schematic diagram of cross-gain modulation [2].

Cross gain modulation employs a nonlinear effect within the SOA called carrier depletion. Normally, the SOA amplifies an optical signal, but when the input gain is too high, after some threshold, there is effectively no output gain. Consider Figure 2, where we have two signals coming into a SOA, our original signal, and a probe signal. The original signal is at λ_s and the probe signal is at our desired wavelength, λ_c . The SOA is configured so that whenever the signal is HIGH, the SOA's carrier will get depleted, so that the output is LOW, so λ_c is LOW. When the signal is LOW, the SOA will output a HIGH, so λ_c is HIGH. **Effectively, we are imprinting the information in λ_s onto λ_c .** Then, we will use a bandpass filter (BPF) at the output of the SOA to filter out λ_c . Note that the information in λ_c will be inverted from the information in λ_s , so usually, we will invert the output to get the correct data. The power of the converted λ_c will have about the same power as the original λ_c .

XGM wavelength converters are very simple to configure. They have high conversion efficiencies, since we directly input the target wavelength into the SOA [3]. Moreover, the SOA has polarization-independent gain, so XGM and XPM are independent of the polarization of the input signal [4]. They are capable of very high bit-rates, since the carrier depletion is a nonlinear effect and occurs in a matter of picoseconds [2].

Looking at Figure 3, we can see that the bit-error rate of XGM can get as low as 10^{-9} , while the eye-diagrams show that the signal still retains its eye, which is desirable [5]. We can cascade two SOAs together to get even better results, as shown.

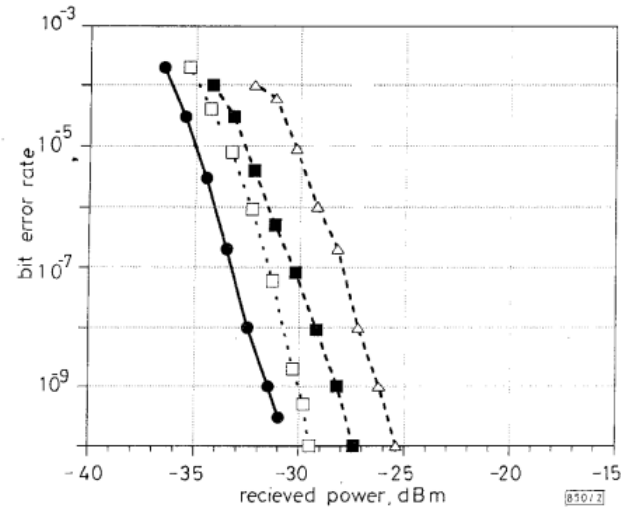


Fig. 2 Bit error rate for 40 Gbit/s wavelength conversion

■ single SOA converter; converted signals at 1548nm ($P_{cw} = 4.5\text{dBm}$)
 △ single SOA converter; converted signals at 1559nm ($P_{cw} = 4\text{dBm}$)
 The input power for the 1554nm signal is $P_m = 5.5$ and 7.5dBm for conversion to 1548 and 1559nm, respectively.
 □ double SOA converter; BER performance for conversion to 1548nm with $P_{cw} = 4\text{dBm}$ and $P_m = 5.5\text{dBm}$
 ● bit error rate for the back-to-back situation at 1554nm

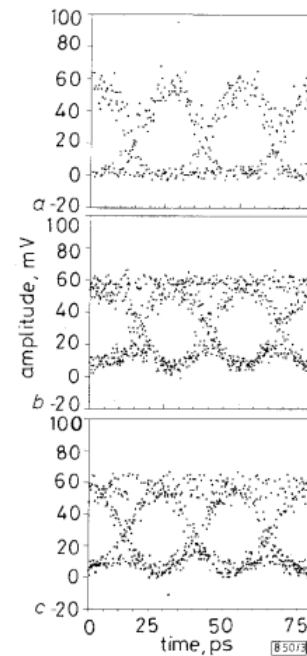


Fig. 3 40 Gbit/s eye-diagrams

a Input signal at 1554nm
 b Converted signal at 1548nm using a single SOA converter
 c Converted signal at 1548nm using a double SOA converter

Figure 3: Bit-error rate and Eye-diagrams of 40 Gbit/s XGM wavelength conversion

The drawbacks of XGM include: the inverted output signal in comparison to the input signal, a large required power to saturate the SOA, and frequency chirping [6].

Cross-Phase Modulation (XPM)

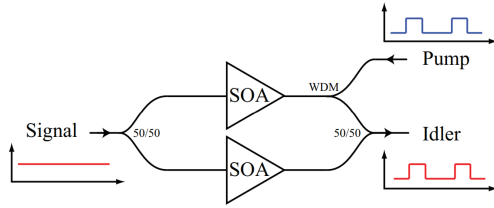


Figure 4: Schematic diagram of cross-phase modulation [7]

Cross-phase modulation (XPM) like XGM, employs carrier depletion in the SOA in order to achieve wavelength conversion. In Figure 4, we now have two SOAs configured in a way to produce a Mach-Zender interferometer (MZI). What's important about the MZI is that it converts phase modulation into intensity modulation. We have our original signal called Pump with wavelength λ_{Pump} , going into one SOA, and another signal with our desired wavelength, λ_{Signal} , going into both SOAs. When λ_{Pump} is HIGH, again, the carriers in the SOA connected to the pump signal will be depleted. In a SOA, when the carriers are depleted, we effectively change the index of refraction of the SOA, which then changes the phase of the incoming signal. And in this MZI configuration, a change in phase causes a change in intensity, so the signal's amplitude is changed. When λ_{Pump} is LOW, we have no carrier depletion, so we get no intensity modulation. **Again, we are effectively imprinting the information in λ_{Pump} onto λ_{Signal}**

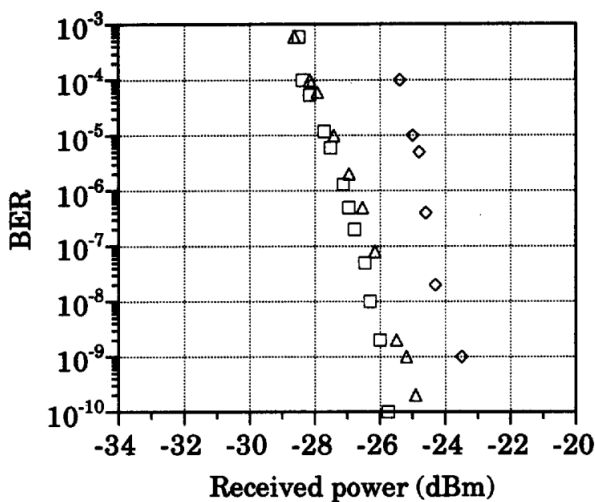


Fig. 5. System experiment at 2.5 Gbit/s. Signal input power: -11 dBm. CW input power: -20 dBm. (□): signal laser (1531 nm) back to back. (Δ): Output signal from MZI converter (1543 nm). (◊): Signal converted by SOA without MZI configuration (1543 nm).

Figure 5: Bit-error rate of 2.5 Gbit/s XPM wavelength conversion [8]

Looking at Figure 5, XPM provides great performance. Similarly to XGM, XPM has excellent conversion efficiency, can support up to 40 Gb/s, and is polarization-independent [9]. What makes XPM better than XGM is that it does not have unwanted frequency chirping. One drawback of XPM is a slightly more complex configuration than XGM with an additional SOA.

Four-Wave Mixing (FWM)

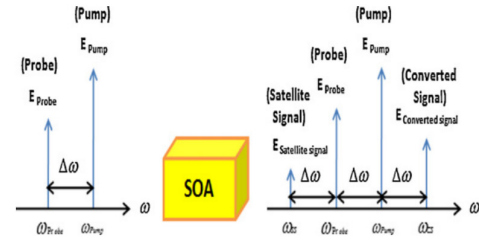


Figure 6: Schematic diagram of four-wave mixing [2]

Four-wave mixing is a nonlinear effect that occurs when two signals beat with each other, creating additional frequencies, as seen in Figure 6. One way to understand this is when we multiply two cosines together, due to the product-to-sum trigonometric identity, we get two additional cosines at different frequencies. In Figure 6, we have two input signals, our original signal with the information, at wavelength λ_{Pump} , and our probe signal at our desired wavelength, λ_{Probe} . When these two signals enter the SOA, they will beat and create two extra wavelengths at the output. **What we can do is place our probe signal at just the right frequency/wavelength, so that at the output, through FWM, we will have a copy of our original signal at our desired wavelength.**

Using FWM, we can achieve error-free wavelength conversion ($\text{BER} < 10^{-12}$) with great signal quality on both 10 Gb/s and 40 Gb/s signals, as shown in the BER plot and eye diagrams in Figure 7 [10]. Like XGM and XPM, FWM is very fast, and can support 40 Gb/s wavelength conversion. FWM is more versatile than XGM and XPM because it can support other modulation formats than ASK, such as PSK and FSK. **This is because in FWM, the amplitude, frequency, and phase are preserved.** In XGM and XPM, they are not preserved.

One drawback of FWM is that it has low conversion efficiency compared to XGM and XPM.

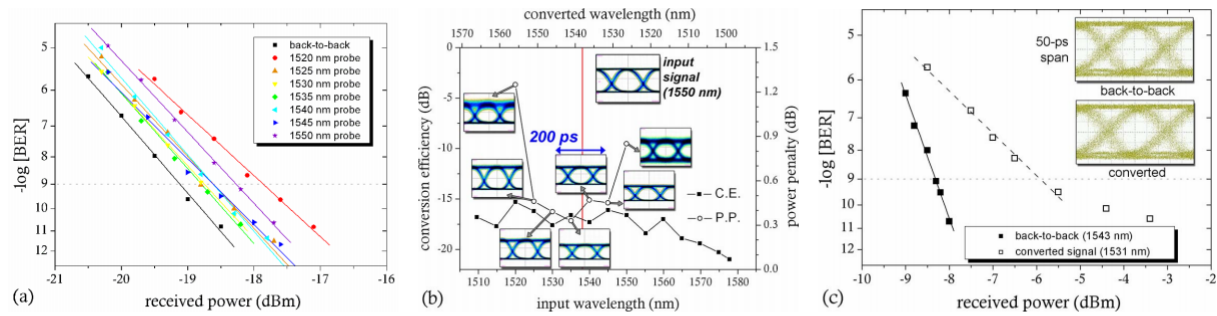


Fig. 2. (a) 10-Gb/s BER curves for converted wavelengths labeled with the input signal wavelength. (b) 10-Gb/s power penalty (○) and conversion efficiency (■) versus wavelength with insets of converted eye diagrams corresponding to each power penalty measurement. The vertical line denotes the location of the pump. (c) 40-Gb/s BER curves with inset eye diagrams before and after wavelength conversion.

Figure 7: Bit-error rate and Eye-diagrams of FWM wavelength conversion of 10 Gb/s and 40 Gb/s signals [11]

Comparison

Out of the three methods for wavelength conversion that I described, FWM is most widely used. This is because it is “transparent” to modulation format, meaning it can support any modulation format, such as ASK, PSK, and FSK. In FWM, we effectively create a copy of the original signal onto another wavelength, preserving its amplitude, phase, and frequency. In contrast, XGM and XPM can only support ASK because the phase and frequency of the original signal is not preserved.

XGM has the highest conversion efficiency out of the three methods, as seen in Table 1. Both XPM and XGM have comparable conversion efficiencies compared to FWM, which is highly desirable.

Summary performances of wavelength converters.

Wavelength converter	Conversion efficiency (dB)
XGM	17
XPM	9.0
FWM	
CW source	-0.7
Spectrum-sliced source	-5.0

Table 1: Conversion efficiencies of XGM, XPM, and FWM [2]

When we want to build a very simple and effective wavelength converter, then XGM is used, since it uses very few components and has a high conversion efficiency. When we want to build an integrated, monolithic wavelength converter, then XPM is used, because unlike XGM, it does not produce chirping, does not produce inverted signals, so it does not need any modification and works right out of the box. When we use other modulation formats other than ASK, we will use FWM.

Comparison

All-optical wavelength conversion (AOWC) employs nonlinear effects to efficiently and quickly convert wavelengths. In this paper, I discussed cross-gain modulation (XGM), cross-phase modulation (XPM), and four-wave mixing (FWM). In application, FWM is more widely used because it is more flexible, but all three methods are used depending on your network architecture’s needs.

AOWC is essential in contention resolution and packet switching, which are becoming increasingly vital solutions towards creating a more efficient optical communications network. AOWC is still improving, and is making great progress. For example, in 2018, we were able to make a 640 Gb/s AOWC [12]! The future truly is bright, and there is much hope for great progress in this field.

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